

Data Visualization Assignment Brief

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Module Details			
Module code:	LDS7004M	Level of Study:	7
Module Leader(s):	Sangita Pokhrel	Credits:	20
Assessment format:	Portfolio (Anonymous)	Method of submission:	Turnitin within Moodle
Deadline or Assessment Period:	9 June 2026, 12Noon	Feedback date and place:	1 July 2026, Turnitin within Moodle
Assessment limits: length, load, word count, etc.	N/A	Component number:	1 of 1
Is this exempt from anonymous marking under the policy?	No	Component weighting:	100%

Assignment Description

Introduction:

In this assignment, you will be creating a data visualization project to communicate insights from a given dataset. Data visualization is an essential skill for effectively presenting and interpreting data. By creating meaningful visual representations of data, you can identify patterns, trends, and relationships that might otherwise be challenging to perceive.

Task:

Your task is to select a dataset and design a data visualization project that effectively communicates key findings and insights. The dataset can be from any domain or topic of your choice, such as social media, healthcare, finance, sports, etc. The objective is to create visually compelling and informative representations of the data.

Requirements:

1. **Dataset Selection:** Choose a dataset that is relevant and interesting to you. Ensure that the dataset is appropriate for creating visualizations and contains enough data points to derive meaningful insights. You can use publicly available datasets from sources like Kaggle, or below data sources.

See the web links for the sample datasets. Choose only one.

- [Kaggle Data Set](#)
- [Inside Airbnb](#)
- [Births, deaths and marriages data](#)
- [National Statistics](#)
- [Data on employment and labour market](#)
- [UCI Machine Learning Repository](#)
- [Data on employment and labour market](#)

2. **Data Cleaning and Preparation:** Clean and pre-process the dataset as necessary. Remove any irrelevant or incomplete data, handle missing values, and format the data appropriately for visualization purposes.

3. **Visualization Techniques:** Utilize a variety of visualization techniques to present the data effectively. Consider using graphs, charts, maps, and other visual elements that best represent the patterns and relationships in the data. Choose appropriate visualization types based on the data characteristics and the insights you want to convey.

4. **Storytelling and Interpretation:** Create a narrative flow in your data visualization project. Clearly explain the purpose of your visualization, the insights you have gained from the data, and the conclusions you draw. Use annotations, captions, or descriptions to provide context and guide the viewer through your visualization.

5. **Aesthetics and Design:** Pay attention to the aesthetics and design principles of your data visualization project. Ensure that the visuals are visually appealing, easy to understand, and well-organized. Use appropriate color schemes, labels, and titles to enhance clarity and readability.

Assignment Description

6. Interactive Elements (optional): If you have the skills and resources, consider adding interactive elements to your visualization project. Interactive features like tooltips, filters, or animations can enhance the user experience and allow for deeper exploration of the data.

7. Documentation: Provide clear documentation of your data visualization project. Include a written report that describes your dataset, data cleaning process, visualization choices, and the insights you have gained. Additionally, provide any code or scripts used to generate the visualizations, along with instructions on how to run them if applicable.

Note: The assessment report with a copy of the code and poster must be converted to a single file before submission. Code to be submitted separately in .ipynb format.

General Consideration

Please be aware that each plot should be fully described in your assessment. But the plot itself must be clear enough to tell a story by itself. You should support your code implementation with written documentation.

You should submit a report summarizing your findings, including tables and charts to support your analysis. Your report should also include a brief discussion of any limitations or recommendations to your analysis.

It is important that the content is underpinned with the inclusion of relevant academic theory, concepts, and models where appropriate, as well as contemporary industrial insights. These should be accurately cited and referenced according to [York St John Harvard Referencing](#) throughout.

Learning Outcomes

You must successfully achieve the following Learning Outcomes to pass this assessment:

This coursework is designed to achieve the following PLOs:

PLOs 7.1-7.5, 7.8, 7.9

7.1 Critically apply skills, techniques, and knowledge from a range of data analysis methods and algorithms for enhancing and solving problems in various domains.

7.2 Develop abstract thinking and design ability to analytically demonstrate concepts relating to data science.

7.3 Use research-based knowledge for the design of experiments, analysis, and interpretation of data to provide valid results.

7.4 Critically evaluate and analyse advanced data science topics, and concepts, and implement them in workplace.

7.5 Identify and implement appropriate programming and software tools to critically analyse big data applications in workplace.

Learning Outcomes

7.8 Critically analyse the data and apply predictive modelling technique in the field of Machine Learning and Artificial Intelligence.

7.9 Critique legal, social, and ethical issues within the field of data science and applicable ancillary sectors, as applied to contemporary research and industry practice.

Advice and Guidance

Assessment Guidance

Deliverables:

1. Written Report: Prepare a comprehensive report that documents your dataset, data cleaning process, visualization techniques used, and the key findings and interpretations from your project. Please make sure to keep the figures, tables, screenshots of the techniques and findings whenever necessary and explain them properly. In the appendix, attach your code with the outputs.

2. Data visualization project: Create a visually engaging Poster that gives an informative data visualization project that effectively communicates insights from the chosen dataset.

3. Code and Instructions: If applicable, include the code or scripts used to generate the visualizations, along with instructions on how to run them.

General Guidance

The work you present should be your own work, and not just copied from others. You can quote from others, but you must say who the author is and use quotation marks or paraphrase. If you do not do so, we will investigate your work for academic misconduct. This is particularly likely if your Turnitin similarity score is high and/or individual matches are high.

If you require support with your study skills, please visit <https://www.yorksj.ac.uk/students/study-skills/>

Assessment Regulations

Please refer to the York St John University Code of Practice for Assessment and Academic Related Matters 2025-26.

We ask that you pay particular attention to the academic misconduct policy. Penalties will be applied where a student is found guilty of academic and/or ethical misconduct, including termination of programme ([Policy Link](#)).

You are required to keep to the word limit set for an assessment and to note that you may be subject to penalty if you exceed that limit. You are required to provide an accurate word count on the cover sheet for each piece of work you submit ([Policy Link](#)).

Advice and Guidance

For late or non-submission of work by the published deadline or an approved extended deadline, a mark of 0NS will be recorded. Where a re-assessment opportunity exists, a student will normally be permitted only one attempt to be re-assessed for a capped mark ([Policy Link](#)).

An extension to the published deadline may be granted to an individual student if they meet the eligibility criteria of the ([Policy Link](#)).

How is this assessment marked?

Your work will be marked according to the assessment instructions provided within this document and the selected Learning Outcomes' (LOs) (see above).

Furthermore, this assessment is marked using the assessment marking criteria or a similar rubric that aligns with the University's Generic Assessment Descriptors (see below).¹ This is to ensure all assessment decisions are comparable regardless of the discipline or mode of assessment.

Please note that you **must** meet the required baseline standards (50 – 59%) which will include the LOs and minimum expectations of the assessment. Further still, you must ensure you meet the requirements of each grade boundary to progress to the next, i.e., you should demonstrate your learning through the standards of the Pass, Merit and Distinction to reach a Distinction (70 – 84%). These standards are designed to scaffold and build your learning to achieve your fullest potential in each criterion being assessed.

¹ A rubric is a type of scoring guide that markers use to set out specific components and expectations for an assignment for their students. It is then used to guide the marking they undertake.

Marking Criteria

Pass Grade Bands (100 – 50) (Learning Outcomes must be met)

Fail Grade Bands (49 – 0) (Learning Outcomes are not met)

Criteria	Deliverables	Marks
1. Thinking Skills	Rigor and extent of the Analysis/discussion. Showing a higher level of critical analysis and insights.	10
2. Research Skills	Learners should provide detailed criticality and evidence of Data import and manipulation. These techniques should follow up with logical and sustained conclusions. ▪ Handling missing or invalid values, if applicable ▪ Correct application of the selected visualization techniques and suitability of the methods.	10
3. Practical, Technical and Professional Learning Skills	Learners should be able to demonstrate technical and professional Exploratory data analysis techniques. Using appropriate statistical methods to explore the data and answer the questions posed in the assignment. Extent of evidence of running the experiments provided in appendices. For example: ▪ Is this a simple dataset or a complex dataset? ▪ Is there sufficient information for the reader to reproduce the results? Is the language used in the report good?	30
	Producing informative and visually appealing plots and tables to support the analysis. ▪ Effective communication of the information contained within the data. ▪ Writing well-organized and well-documented code that follows best practices and is easy to read and understand ▪ Demonstrating proficiency with Python libraries and programming concepts.	25

	Demonstrating critical thinking and creativity in identifying and interpreting trends and patterns in the poster presentation. Justification for the selection of chosen visualization techniques.	15
4. Communication and Professional Conventions	Citations are provided in Harvard's style and the report is well organized and presented. Producing a clear and concise written report that summarizes the analysis and findings. ▪ Presenting the results in a visually appealing and informative way. ▪ Communicating technical concepts in a way that is accessible to a non-technical audience.	10

Level 7 GAD Descriptor for Assessment Matrix

Criteria	Pass (50 – 59)	Merit (60 – 69)	Distinction (70 – 84)	Distinction (85 – 100)	Borderline Fail (45 - 49)	Fail (30 - 44)	Fail (0 - 29)	Marks Out of 100
Adjectives	Satisfactory	Good	Excellent	Exemplary	Not Successful	Unsuccessful	Fail	
Thinking Skills	The work attempts to apply advanced thinking skills methodically to material from the taught content, demonstrating a satisfactory understanding of the topic. Some evidence of critical analysis is present but lacks depth.	The work systematically applies advanced critical thinking skills, effectively identifying problems and proposing appropriate solutions. The analysis is more comprehensive and insightful.	The work offers significant new insights into the research topic, demonstrating a deep understanding and advanced critical thinking. Problems and solutions are well-articulated and innovative.	The work creates new insights or hypotheses, demonstrating cutting-edge thinking and professional awareness. It shows exceptional application of advanced thinking skills.	The work shows some application of thinking skills but lacks sufficient depth and breadth. Core content needs a more comprehensive application.	The work applies minimal thinking skills, with many gaps and misunderstandings in core content. The analysis is limited in scope and depth.	The work does not apply the required thinking skills. It is limited by significant misunderstandings and lacks depth in addressing the taught content.	10
Research Skills	The work attempts to use research methods correctly but lacks depth. It shows some ability to solve problems and answer research questions.	The work successfully utilizes research methods and demonstrates a solid understanding of their application. It effectively generates insights and answers research questions.	The work employs a systematic and thoughtful approach to research, showing strong competency and a thorough understanding of methods. It contributes significantly to the field.	The work demonstrates mastery in research design and methodology, producing new knowledge and adhering to professional standards. It articulates the researcher's position clearly and effectively.	The work partially applies research methods but lacks coherence in addressing problems or research questions. It needs improvement in method relevance and application.	The work fails to apply necessary research methods effectively. There is a lack of clarity in how methods are relevant to the research questions.	The work does not address required research methods or attempt to solve research problems. There are significant misunderstandings and misapplications.	10

Practical, Technical and Professional Learning Skills	The work demonstrates basic exploratory data analysis techniques and uses statistical methods to answer assignment questions. Evidence of experiments may be limited, and the dataset analysis may be simple.	The work applies a range of exploratory data analysis techniques effectively and uses appropriate statistical methods. It provides sufficient detail for reproducibility and clear documentation.	The work shows proficient use of complex exploratory data analysis techniques and statistical methods. The experiments are well-documented, and the dataset analysis is thorough and sophisticated.	The work demonstrates advanced technical skills with complex datasets, providing detailed and reproducible results. The analysis is both thorough and innovative, adhering to high professional standards.	The work shows limited use of exploratory data analysis techniques and statistical methods. There may be gaps in documentation or reproducibility of results.	The work applies few exploratory data analysis techniques effectively. The use of statistical methods and documentation may be inadequate, impacting the reproducibility and clarity of results.	The work fails to apply relevant exploratory data analysis techniques and statistical methods. Documentation is poor, and results are not reproducible or clear.	30
Practical, Technical and Professional Learning Skills	The work produces informative plots and tables but may lack visual appeal. Communication is adequate but could be more effective. Code quality may show some issues with organization or documentation. Justification for visualization techniques is present but may lack depth	The work produces visually appealing plots and tables. Communication is effective, and code is well-organized and documented, showing a good understanding of best practices. Justification for visualization techniques is clear and relevant.	The work creates highly informative and visually compelling plots and tables. Communication is clear and effective, with well-documented and organized code demonstrating proficiency with Python libraries and programming concepts. Visualization techniques are well-justified	The work excels in producing exemplary plots and tables, with exceptional communication of data insights. Code quality is top-notch, adhering to best practices and demonstrating advanced proficiency in Python libraries and programming concepts. Visualization techniques are expertly chosen	The work shows some effort in producing informative plots and tables but lacks in visual appeal and clarity. Code quality and communication need improvement. Justification for visualization techniques may be insufficient.	The work produces plots and tables that are not fully informative or visually appealing. Code may be poorly organized or documented, affecting clarity and communication. Justification for visualization techniques is weak or lacking.	The work fails to produce useful plots or tables. Communication is unclear, and code quality is poor, with significant issues in organization and documentation. Visualization techniques are not justified or poorly selected.	25

			and effectively applied.	and justified, exceeding industry standards.				
Practical, Technical and Professional Learning Skills	The poster demonstrates a basic level of creativity and information presentation. Insights are present but not particularly deep or innovative. The poster communicates key points adequately, though it may lack visual appeal or clarity in some areas. The work demonstrates some critical thinking and creativity in data interpretation.	The poster shows effective creativity and clear presentation of information. Insights are more developed and relevant. The visual elements of the poster are well-organized and contribute to a better understanding of the data, enhancing overall communication. The work shows good critical thinking and creativity in identifying trends and patterns.	The poster is highly creative and presents information in an engaging and insightful manner. The insights provided are thorough and well-supported. Visual elements are compelling and professionally designed, significantly enhancing the effectiveness of the presentation. The work excels in critical thinking and creativity, offering insightful interpretations of trends and patterns.	The poster excels in creativity and presentation, offering innovative and original insights. It is exceptionally well-organized and visually striking, demonstrating a high level of proficiency in data visualization. The presentation effectively communicates complex ideas clearly and professionally. The work demonstrates exceptional critical thinking and creativity, leading to innovative interpretations of data.	The poster shows some effort in creativity and presentation but lacks coherence or visual appeal. Insights are superficial or poorly articulated. The communication of information is unclear or incomplete, impacting overall effectiveness. The work shows limited critical thinking and creativity.	The poster lacks creativity and fails to present information effectively. Insights are minimal or unclear. Visual elements are poorly designed, making the poster difficult to understand and less effective in conveying the data. The work displays minimal critical thinking and creativity.	The poster fails to demonstrate creativity or effective presentation. There is a significant lack of meaningful insights, and the visual elements are inadequate or missing. The overall communication of information is ineffective and poorly executed. The work shows no critical thinking or creativity in interpreting data.	15

Communication and Professional Conventions	Adheres to all professional and academic conventions, including proper citations and report organization. The report excels in clarity and presentation, with exemplary communication of complex ideas, impeccable Harvard-style citations, and ethical use of technologies. It is well-structured and engages a broad audience effectively. Pass (50-100%)	Fails to adhere to professional conventions with major errors or inconsistencies. The report is unclear and poorly organized, with significant issues in citation style and communication. Technologies are not used effectively. Fail (0-49%)	10
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